



**Verified Carbon
Standard**

YILMAZLAR BIOGAS POWER PLANT PROJECT



Document Prepared by Life İklim ve Enerji Ltd. Şti.

Project Title	Yılmazlar Biogas Power Plant Project
Version	01
Date of Issue	24-February-2023
Prepared By	Life İklim ve Enerji Ltd. Şti.
Contact	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara, Tel: 0090 312 481 2142, info@lifeenerji.com, www.lifeenerji.com

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	4
1.3	Project Eligibility	5
1.4	Project Design	5
1.5	Project Proponent	6
1.6	Other Entities Involved in the Project	6
1.7	Ownership.....	6
1.8	Project Start Date	6
1.9	Project Crediting Period	7
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	7
1.11	Description of the Project Activity.....	8
1.12	Project Location	11
1.13	Conditions Prior to Project Initiation	13
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks.....	14
1.15	Participation under Other GHG Programs	15
1.16	Other Forms of Credit.....	15
1.17	Sustainable Development Contributions	16
1.18	Additional Information Relevant to the Project	21
2	SAFEGUARDS	22
2.1	No Net Harm	22
2.2	Local Stakeholder Consultation	22
2.3	Environmental Impact	23
2.4	Public Comments	24
2.5	AFOLU-Specific Safeguards	24
3	APPLICATION OF METHODOLOGY.....	24
3.1	Title and Reference of Methodology	24
3.2	Applicability of Methodology.....	25
3.3	Project Boundary	30
3.4	Baseline Scenario	32

3.5	Additionality	36
3.6	Methodology Deviations	36
4	ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	38
4.1	Baseline Emissions	38
4.2	Project Emissions	42
4.3	Leakage.....	48
4.4	Estimated Net GHG Emission Reductions and Removals.....	49
5	MONITORING	54
5.1	Data and Parameters Available at Validation	54
5.2	Data and Parameters Monitored.....	54
5.3	Monitoring Plan.....	60
6	ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS	74
6.1	Data and Parameters Monitored.....	74
6.2	Baseline Emissions	74
6.3	Project Emissions	77
6.4	Leakage.....	77
6.5	Net GHG Emission Reductions and Removals.....	77
	APPENDIX X: <TITLE OF APPENDIX>	78

1 PROJECT DETAILS

1.1 Summary Description of the Project

Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi (hereinafter mentioned as Project Owner) has constructed a cattle farm with the name of Kayı Farm located in Derebent Village of Tavşanlı District of Kütahya Province in Türkiye. The Project Owner has been operating the farm since 2013. As a result of farm activities, some amount of manure is being generated. To treat this waste, Project Owner invested in Yılmazlar Biogas Power Plant Project (hereinafter mentioned as the Project), and the construction of the plant was completed and started operation in June 2021. Thanks to the Project, the manure resulting from the farm activities can be disposed of safely and environmentally friendly, and clean energy can be generated.

The Project has been designed to have a lifetime of 49 years in accordance with generation license. It was planned that cattle manure from Kayı Farm and chicken litter from nearby farms would be mixed and digested to produce biogas by anaerobic digestion and then converted to electricity biogas power plant. The remaining digestate from the anaerobic digesters is treated further to produce organic fertiliser.

On 20 May 2020, EMRA granted the Project activity with a Generation License of 1.560 MWe and 11,700 MWh/year for 49 years. The license is granted for the Plant to have 1 gas engine with a capacity of 1.560 MWe. The commissioning date of the plant is 27 June 2021. The construction of the power plant was started in February 2020 after signing the Construction Agreement with the contractor company and then the Plant started generating electricity on 27 June 2021 by connection agreement with EPIAS.

Prior to the start of the project activity (in the baseline scenario) manure from the farms were being washed out from underneath the animal barns and stored in anaerobic lagoons which allowed the manure to decay and emit methane to the atmosphere along with several other toxic gases. Such applications also bear the risk of lagoon overflow due to increased rainfalls, strong Biogas or improper construction.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 Renewable Energy

Sectoral scope: 13 Waste Recovery and disposal

Type I: Renewable Energy Projects

Type III: Other project activities

The Project is neither an AFOLU nor a grouped project

1.3 Project Eligibility

The scope of the VCS Program includes:

- 1) The six Kyoto Protocol greenhouse gases: The project is expected to avoid Methane (CH₄) emissions from the anaerobic animal manure management system in the baseline scenario, which will be captured and destroyed in the project scenario as well as renewable energy generation. Thus, the project applicable to this scope.
- 2) Ozone-depleting substances: Not Applicable.
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: Not Applicable.
- 4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The applied methodologies AMS-III.D (Version 21.0) and AMS-I.D. (Version 18.0) are methodologies approved under CDM Program, which is a VCS approved GHG program.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

The biogas generated from the anaerobic digesters is recovered and utilized as an energy source within the project boundary. Apparently, the project does not generate GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

Furthermore, the project does not belong to the project activities excluded in Table 1 of VCS Standard 4.4.

Thus, the project is eligible under the scope of VCS program.

1.4 Project Design

- ☒ The Project includes a single location or installation only
- ☐ The Project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The Project is a grouped project

1.5 Eligibility Criteria

The Project Activity is not a grouped project. The Project Activity belongs to one project proponent. In the scope of this project activity, only one location will be considered for emission reductions.

1.6 Project Proponent

Organisation name	Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi
Contact person	Mr Ramazan Fidan
Title	Biogas Plant Operation Head
Address	Tunçbilek Karayolu 7. Km Tavşanlı/Kütahya/Turkey
Telephone	+90 (274) 600 02 00
Email	r.fidan@yilmazlarmadencilik.com.tr

1.6 Other Entities Involved in the Project

Organisation name	Life İklim ve Enerji Ltd. Şti.
Role in the Project	Consultant for Carbon Certification
Contact person	Mr. Ramazan Aslan
Title	General Manager
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara
Telephone	+90 (312) 481 21 42
Email	ramazan.aslan@lifeenerji.com

1.7 Ownership

Ownership of the Yılmazlar Biogas Power Plant project belongs to Yılmazlar Madencilik San. ve Tic. Ltd. Şti., and all emission reductions/removals are granted to the company. The tax plate as proof of operating certification of the company given by the relevant tax office of the host country is provided to the validating VVB as proof of the Title.

1.8 Project Start Date

The project start date is determined by the earliest commissioning date of the biogas engines. The start date of the project activity is 27-June-2021, i.e. the provisional acceptance protocol of the engine.

1.9 Project Crediting Period

The Project crediting period for this Project is 7 years, renewable twice.

First Crediting Period Start Date: 27 June 2021

First Crediting Period End Date: 26 June 2028

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the Project are:

☐ <20,000 tCO₂e/year

☒ 20,000 – 100,000 tCO₂e/year

☐ 100,001 – 1,000,000 tCO₂e/year

☐ >1,000,000 tCO₂e/year

Project Scale	
Project	X
Large Project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
27 th June 2021 – 31 st December 2021	20,626
01 st January 2022 – 31 st December 2022	40,257
01 st January 2023 – 31 st December 2023	40,257
01 st January 2024 – 31 st December 2024	40,257
01 st January 2025 – 31 st December 2025	40,257
01 st January 2026 – 31 st December 2026	40,257
01 st January 2027 – 31 st December 2027	40,257
01 st January 2028 – 26 th June 2028	19,632
Total estimated ERs	281,800
Total number of crediting years	7
Average annual ERs	40,257

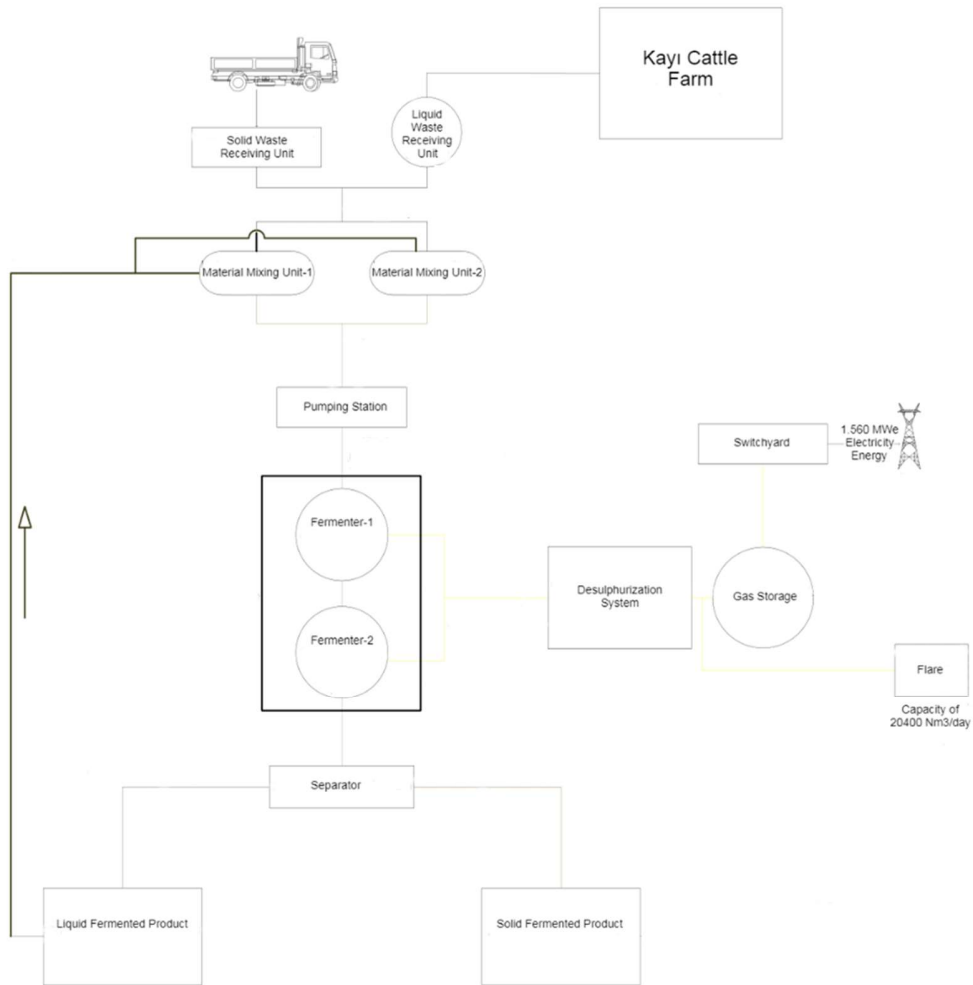
1.11 Description of the Project Activity

As mentioned before, two sources of waste are being digested in the Yılmazlar Biogas Power Plant. Cattle manure comes from the Kayı Farm in which Yılmazlar Biogas Power Plant is, which belongs to the same project owner as the plant, by pipes. On the other hand, chicken litter is brought from poultry farms around by trucks.

Yılmazlar Biogas Power Plan is designed to include units listed below:

- Solid Waste Receiving Unit
- Liquid Waste Receiving Unit
- Material Mixing Unit
- Anaerobic Digesters
- Desulphurization and Gas Storage
- Separator Units
- Gas Engine and Flare Units

Process diagram of the plant is given in the figure below.



Solid Waste Receiving Units

Chicken litter supplied from other farms is first taken to the solid waste receiving unit. The solid waste receiving unit is made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material, so it has a leakproof base and closed top and sides.

The purpose of the solid waste receiving unit is to empty the chicken litter of the vehicles arriving at the facility as soon as possible and to store the wastes appropriately throughout the year. Solid wastes in the unit are taken to the material mixing unit in the amount to be used daily.

The area of the solid waste receiving unit is 300 m². The base of the unit was designed to collect leachate from the bottom to prevent contamination of any water body, and collected leachate is directed to the material mixing unit.

Liquid Waste Receiving Units

All liquid waste generated and received by the plant is animal manure produced in Kayı Farm where Yılmazlar Biogas Power Plant is actually in. Liquid wastes are collected from the farm and pumped to the power plant by pipeline.

Liquid wastes are firstly taken to the liquid waste receiving unit. The liquid waste receiving unit is made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material, so that it has leakproof base and closed top and sides.

Solid and liquid wastes accepted by the plant and is mixed in order to provide suitable moisture for digestion. Material mixing units are made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material so that it has a leakproof base and closed top and sides.

Mixing Unit

Solid and liquid waste is mixed daily in two mixing units. The aim of having two material units is when one unit is in mixing process, the other one pumps the waste into digesters. Therefore, when one unit gets empty, the other will be ready to pump. Consequently, the non-stop feeding of digesters is maintained.

Each material mixing unit has an 84 m³ capacity (168 m³ total). To have a homogenous mixture, each of them has a submersible mixer. Moreover, they have biofilters to prevent odor.

Anaerobic Digesters

There are two anaerobic digesters in the biogas plant with 6430 m³ capacity each. The reactors are made from reinforced concrete, leak-proof, and designed to have a self-heating and mixing system. The waste mixture that has suitable moisture is digested by microorganisms under anaerobic conditions and converted into carbon dioxide and methane gas. In the digester, methane gas is produced under mesophilic conditions. The methane gas produced is withdrawn by blowers from the digester and delivered to the biogas engine unit to obtain electrical energy. Hot water pipes placed inside the digesters provide automatic control of the temperature of the digester to keep it constant with continuous hot water circulation.

Also, mixers inside the digesters are activated according to the time parameters determined in the Plant's automation system to keep the mixture homogenous.

Before starting the project activity (in the baseline scenario), manure from the farms was being washed out from underneath the animal barns and stored in anaerobic lagoons, which allowed the manure to decay and emit methane into the atmosphere along with several other toxic gases. Such applications also bear the risk of lagoon overflow due to increased rainfalls, strong Biogas or improper construction.

Energy requirements for the Plant will be supplied from the national grid, and a generator is ready to use in an emergency electricity need.

The Project is estimated to generate a net 11,700 MWh/y of electricity in accordance with generation license, while the biogas production capacity will be around 6,275,335 m³/y

Desulphurisation Unit and Gas Storage

Hydrogen sulphide, a highly corrosive acidic gas contained in the biogas, and water vapour are also eliminated in the desulphurisation unit employed in the biogas plant. Some amount of hydrogen sulphide (H_2S) in the biogas is reduced by addition of oxygen to the digesters. After that product gas is taken to desulphurisation unit and hydrogen sulphide is reduced to desired level. The produced biogas is stored in double membrane inflatable roofs and then accelerated by a blower system to the gas engine.

The biogas stored in the gas balloon is sucked from this area by means of gas blowers. The biogas, which is compressed and pressurized by the gas blowers, is transported to the inlet line of the biogas engine system at the desired pressure.

Gas Engine and Flare Unit

The gas engine unit has one electricity-generating engine with a capacity of 1.56 MWe. Biogas energy fed to gas engines will be converted into electricity with 42.1% efficiency in accordance with feasibility report. All the electricity produced is supplied to the grid, which also meets the electricity consumed by the Plant. There is also a flare unit with a capacity of 850 m³/hr in the biogas plant for emergencies where the engines don't combust produced biogas. The combustion temperature of the flare is around 1200°C.

Dewatering (Separator)

Two separators are employed to separate solid and liquid phases of digestate. Separators can separate digestate up to 25% dry matter and 95% liquid matter. The area of the separator unit is 50 m².

1.12 Project Location

The Project is located in Derbent Village of Tavşanlı District of Kütahya Province in Turkey. Total area of the project is 24.330.18 m². Derbent village is 1250 meters north of the project, Yeniköy village is 2120 meters south, and Demirbilek Village is 3500 meters west which are the nearest settlements.

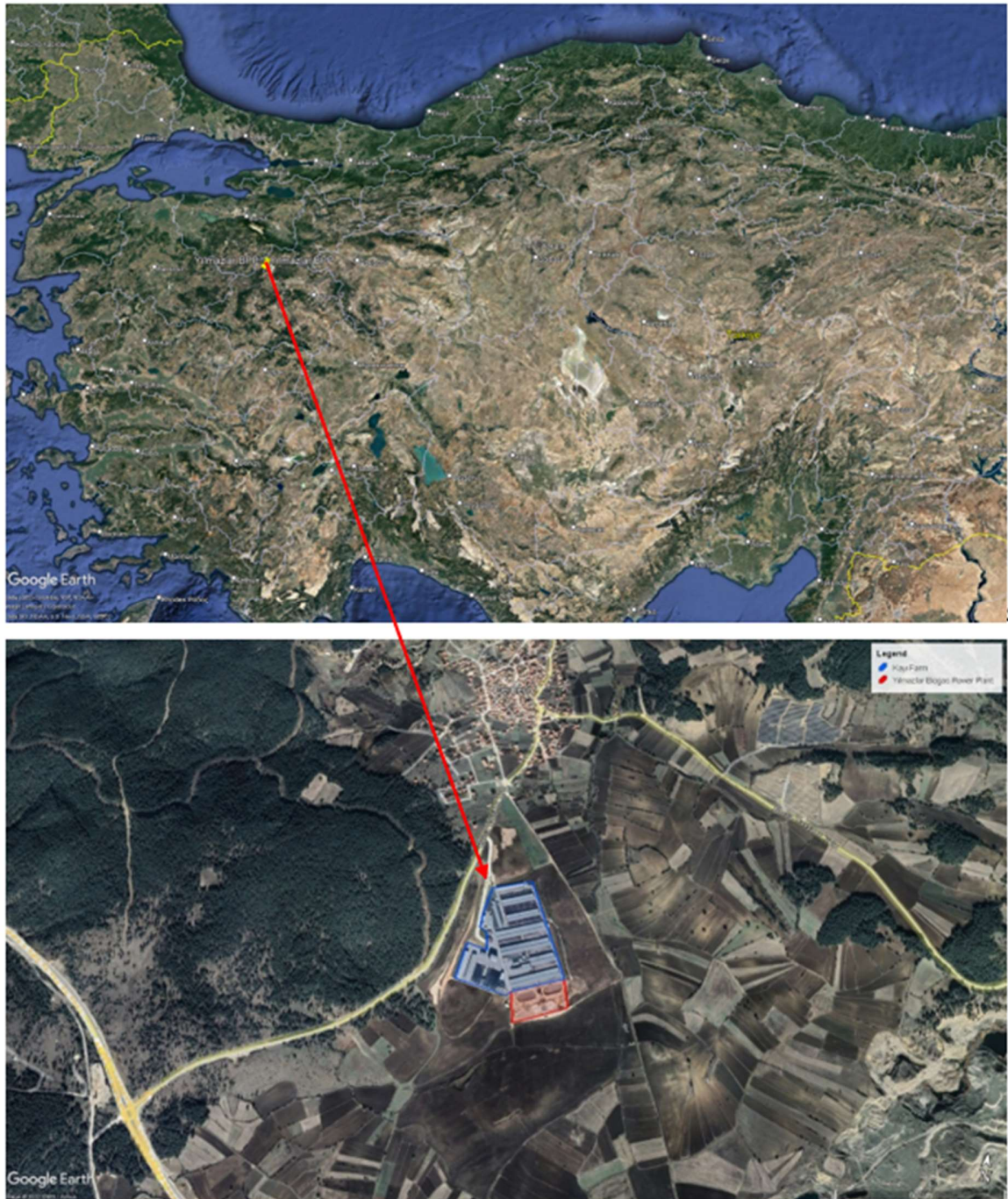


Figure 1: Satellite View of Yilmazlar Biogas Power Plant

Corner coordinates of the plant is given in the table below.¹

¹ Generation License

Point No	E	N
1	702062,690	4389355,080
2	701996,380	4389321,060
3	701992,300	4389329,030
4	701965,100	4389315,090
5	701969,140	4389307,200
6	701902,790	4389273,200
7	701920,310	4389236,870
8	701931,410	4389242,590
9	701945,580	4389214,930
10	701950,030	4389217,210
11	701954,630	4389208,440
12	701974,990	4389218,870
13	701980,440	4389208,230
14	702010,700	4389223,740
15	702015,850	4389213,680
16	702042,510	4389227,330
17	702037,340	4389237,430
18	702067,570	4389252,680
19	702043,500	4389299,820
20	702081,540	4389319,240

1.13 Conditions Prior to Project Initiation

The alternative scenarios of animal manure in the absence and prior to the project activity include:

- Electricity would be generated in existing grid-connected electricity plants
- The animal manure would be treated in the anaerobic lagoons

Before the project activity, cattle and chicken litter generated from cattle breeding and poultry farms were left to decay in anaerobic lagoons. Therefore, greenhouse gases generated from the manure would be directly released into the atmosphere. Moreover, the electricity generated by the Project would be replaced by the power plants connected to the national grid, and fossil fuel-fired boilers would supply the thermal energy.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

Turkish legislation on manure management concentrates on control of the manure waste, and its potential threat on air, water and soil pollution.

The related legislation is presented below:

Relevant Law and Regulations for the Project Activity	Comment
“Environmental Law ² ” dated 26/04/2006 numbered 2872	The law prohibits the discharge, storage, transfer, and disposal of any kind of waste that does not meet the requirements according to Turkish regulations and standards. It includes the “polluter pays principle” and ensures that the producers take necessary measures to minimise waste. The law doesn’t mandate any alternative treatment processes for the animal manure management.
Communiqué on Fermented Product Management by Mechanical Separation, Biodrying, Biomethanization dated 10/10/2015 numbered 29498	The purpose of this Communiqué is to ensure the management of biodegradable wastes without harming the environment and human health, to reduce the amount to be disposed of in landfills, to the technical criteria of mechanical separation, biodrying and biomethanization facilities with material or energy recovery facilities, to the management of the fermented product obtained in biomethanization facilities.
Regulation on Control of Odor Generating Emissions	The purpose of this Regulation is to regulate the administrative and technical procedures and principles for the control and reduction of odor-causing emissions.
Electricity Market Law ³ dated 20/02/2001 and numbered 4628	The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market. The regulation doesn’t mandate any alternative treatment processes for electricity production.
Law on Utilisation of Renewable Energy Resources for the Purpose of Generating	The purpose of this Law is to expand the utilisation of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to

² See: <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5>

³ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf>

Electricity Energy ⁴ dated 10/05/2005 and numbered 5346	increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the environment and to develop the related manufacturing sector for realising these objectives. The Law brings an incentive of 13.3 \$ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration systems etc. The regulation does not mandate any alternative treatment processes for renewable electricity production.
--	---

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has neither been registered nor seeking registration under any other GHG programs. The Project is seeking registration only in the VCS program.

1.15.2 Projects Rejected by Other GHG Programs

No, the project activity didn't apply for registration in any other GHG program and was not rejected by any other GHG programme.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the Project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

There are no emissions trading programs or any other mechanism in Turkey that includes GHG allowance trading in Turkey; subsequently, the Project does not fall under any such mechanisms.

1.16.2 Other Forms of Environmental Credit

Has the Project sought or received another form of GHG-related credit, including renewable energy certificates?

⁴ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf>

☐ Yes☒ No

The Project has not sought or received any other form of GHG-related environmental credits.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

SDG 13 Climate Action

The Project Activity results in emissions reduction by capturing the methane in cattle and chicken litter and converting it to electricity in the biogas power plant. Without the Project Activity, the manure generated in the farms would be left to decay in an anaerobic lagoon producing methane gas. With the Project Activity, the collected animal manure is being used in the digesters to produce biogas and energy, avoiding the release of methane into the atmosphere along with several other toxic gases. By avoiding methane emissions, the project activity will result in an average emission reduction of 40,257 tCO₂e annually. Additionally, in the biogas plant, captured methane is converted to electricity, where the generated electricity is supplied to the grid. Without the Project Activity, the electricity generated would be supplied from the national grid where thermal power plants are connected, causing more GHG emissions.

SDG 7 Affordable and Clean Energy

The Project Activity results in renewable energy generation by digesting animal manure in an anaerobic environment to produce electricity. The baseline scenario for using gas generated from an anaerobic digester (biogas) is that the electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and/or by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of fossil fuel resources in increasing energy demand of the country; has also increased significantly in recent years. In the absence of the proposed project activity, the same amount of electricity is required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus increasing GHG emissions.

The clean energy generated by the Project Activity is calculated based on the electricity generated per annum. The Project is expected to generate 11,700 MWh of clean energy annually and contributes to the share of renewable energy sources.

In order to estimate the outcome of SDG 7, Indicator 7.2.1, and “Renewable energy share in the total final energy consumption” of SDG Target 7.2: “By 2030, increase the share of renewable energy in the global energy mix substantially” can be employed. The project activity will increase renewable energy share in the country’s energy mix.

There is no electricity generation in the baseline scenario in which Yılmazlar BPP is not being implemented.

The Project's contribution to SDG 7 is determined via a simple calculation that provides the overall renewable energy generation fed into the grid. This approach is based on the following calculation:

The quantity of net electricity generation supplied by the project plant to the grid in the related year has been calculated by subtracting "Electricity consumption from the grid (MWh)" from "Electricity supplied to the grid (MWh)"

Calculation of Electricity Generation of Yilmazlar BPP:

Net Electricity Generation of Yilmazlar BPP = Electricity supplied to the grid (MWh) - Electricity consumption from the grid (MWh)

SDG 8 Decent Work and Economic Growth

Turkey has a high rate of unemployment. The Project Activity employs more than 16 people, more than half local, and all employee working in the Plant is trained at least once a year. Without this Project, there would be no employment and no training. Therefore, it would not be possible to contribute to the project region's economy and the people's education without the project activity. Health and safety (HSE) training certificates for all employees are provided, and all employees are to be trained according to the requirements. The Project Activity promotes inclusive and sustainable economic growth, employment and decent work for all and contributes to the decrease in the unemployment rate.

1.17.2 Sustainable Development Contributions Activity Monitoring

SDG13: Climate Action

The emission reduction parameter can analyse this contribution. The Project's annual average emissions reduction is estimated to be 40,257 tCO₂e for this crediting period.

SDG8: Decent Work and Economic Growth

The Project has employed 16 workers. The employment of local people with the necessary technical qualifications for the required post has been the priority and enhanced by all project activities during the construction and operation of the Yilmazlar BPP. As a result, increased job opportunities and project business activities have partially eliminated local poverty and unemployment. Construction materials for the foundations, cables, and other auxiliary equipment have been preferentially sourced locally. Moreover, as a contribution of the Project to the region's welfare, the quality of the Electricity consumed in the area has been increased by local electricity production, contributing to decreasing distribution losses.

Moreover, project activity improves the quality of employment by training the employee. Thus, a series of training will be organised with different focuses such as health and safety, environmental management and technical capacity building within the power plant.

SDG7: Renewable Affordable and Clean Energy

The Project's gross annual renewable electricity generation has been estimated to be 81,900 MWh during this Crediting Period.

In the baseline scenario, cattle manure generated from cattle breeding was left to decay in anaerobic lagoons. Therefore, greenhouse gases generated from the manure would be released into the atmosphere. The baseline scenario for the use of gas generated from a biogas power plant is that the electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of renewable resources⁵; has decreased significantly in recent years. There is an increasing demand for electricity, which is expected to continue in the foreseeable future. In the absence of the proposed project activity, the same amount of electricity is required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus increasing GHG emissions.

⁵ Please see <https://webapi.teias.gov.tr/file/1bfb54a8-26c8-4e65-b354-f47044a1bf6b?download>

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	SDG Target number	Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator	Indicate the Project's contribution to the SDG Indicator (implemented activities to increase or decrease)	Brief description of the quantifiable impact of the Project's activities related to the SDG indicator, during the monitoring period.	Brief description of the cumulative quantifiable impact of the Project's activities related to the SDG indicator, over the project lifetime.
1)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	the annual average emission reduction of the Project is estimated to be 40,257 tCO ₂ e	the annual average emission reduction of the Project is estimated to be 40,257 tCO ₂ e
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities	Implemented activities to increase	During the crediting period, the Project employs a total of 16 workers.	Since the operation start date of the Plant, the Project has employed a total of 16 workers.

3)	8.8	8.8.2 Level of national compliance with labour rights (freedom of association and collective bargaining) based on International Labour Organization (ILO) textual sources and national legislation, by sex and migrant status	Implemented activities to increase	During the crediting period, the Project provides a total of 16 training to all employee	Since the operation start date of the Project, the Project has provided a total of 16 training to all employee
4)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	the annual average electricity generation of the Project is estimated to be 11,700 MWh/year.	the annual average electricity generation of the Project is estimated to be 11,700 MWh/year.

1.18 Additional Information Relevant to the Project

Leakage Management

The floor of the waste reception unit is made of 30 cm thick, C30/37 concrete and non-flammable material to ensure its impermeability. A separate collection mechanism is constructed at the base of the waste reception unit to prevent the waste from contacting or leaking with groundwater, sewage or surface water. The ground is sloped appropriately to collect the leachate in the waste reception area. Also, an impervious concrete floor is applied for leachate on the Material Recovery Facility floor, and grate channels are used to collect the leachate. Collected leachate is given to material mixing unit.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

There is no further information that may have a bearing on the Project's eligibility, the net GHG emission reductions or removals, or the quantification of the Project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

The Ministry of Environment and Urbanization exempted the project from EIA on 28.01.2021. This decision is given in accordance with the scale of the project, and this decision does not exempt the project from all regulations and legislations. Moreover, the project identification document was prepared and approved by the Ministry. Considering that, it is possible to claim that there is no net harm related to the project.

2.2 Local Stakeholder Consultation

A stakeholder meeting was conducted on 15/11/2022 for Yılmazlar BPP Project. The meeting location was chosen as the Kayı Farm in Derbent, where the plant is located in. The meeting was held in the farm's meeting room, which the local people could easily access. The announcement of the stakeholder meeting was sent by post on 02/11/2022 to the Kutahya Provincial Directorate of Environment and Urbanization, Tavşanlı Municipality as well as NGOs related with environmental issues by mail. Subsequently, village mukhtars were also contacted by telephone to provide further information about the meeting. There was participation in the meeting from the local residents, the deputy Mayor of Tavşanlı Municipality, employees of the Plant, the Project Proponent and the mukhtars of the villages. During the Stakeholder Consultation Meeting, the participants were given detailed information about the planned activity through a visual presentation.

The mechanism for ongoing communication was established with the support of the Mukhtar for stakeholders to forward their requests or concerns regarding the project activities. A logbook for receiving comments from the locals was left in Mukhtar's office. In Turkey, mukhtars are the most relevant persons in the villages for establishing a grievance mechanism. Their offices are some of the few places the locals visit very often for different purposes, such as requesting some official letters. Whenever a problem arises in the village, as the first thing locals go to the mukhtars office to express their concerns to find solutions and reflect on their issues with the Mukhtar. It has been decided that the Mukhtar will also take an active role in monitoring the participants' requests so that a grievance mechanism situating the Mukhtar at the centre was established for this project activity. Mukhtar was provided with all the contact information of the responsible persons for the project activity. Whenever a complaint arises from the local people or Mukhtar himself is concerned about the project activity, he can directly contact the relevant responsible persons.

In general, local stakeholders were supportive of the project activity. Most of the local stakeholders thought the Project would help improve the livelihood of local people and promote local economic development without much adverse environmental impact. Therefore, the project

implementation is regarded as beneficial by most of the stakeholders. No negative comments from the stakeholders were received during this meeting.

2.3 Environmental Impact

Within the scope of project identification document (PID) submitted to Ministry of Environment, Urbanization and Climate Change key environmental impacts were assessed.

Air Pollution

In accordance with the approved PID, emission from biogas utilisation will be measured and reported to the Ministry. The project owner committed that emissions will not exceed the threshold given in "Industrial Air Pollution Control Regulation". There is no odor problem expected from digested product since CH₄ that causes odor will be used in gas engines. The odor can be expected only in the waste material receiving section of the facility. In order to minimize the odor in the incoming waste, it will be ensured that the waste reception units are completely closed.

Water Pollution

Most liquid fertiliser obtained from the biogas plant is recycled back into the system. The remaining liquid fertiliser product formed in the Biogas Plant is supplied to villagers as liquid fertiliser. At every stage of the Project, the provisions of the "Water Pollution Control Regulation", which entered into force after being published in the Official Gazette dated 31.12.2004 and numbered 25687, will be complied with. Domestic wastewater generated by the plant is vacuumed by local municipality.

Soil Pollution and Solid Waste

In order not to cause soil pollution, production and final product storage activities will be on an impermeable concrete floor. The waste storage area will be built on an impermeable concrete floor. The Activity Preliminary Information Form will be submitted to the Provincial Directorate of Environment, Urbanization and Climate Change, which will be made over the Contaminated Site Information System, within 30 days after the facility is operational. In the event of a situation requiring renewal of the notification, it will be renewed.

Domestic solid wastes, which will be formed from the social activities of the personnel to be employed in the project, will be collected in a sealed and impermeable garbage barrel that will not emit odor during the construction and operation phases. Domestic solid wastes, the relevant articles of the Waste Management Regulation, published in the Official Gazette dated 02.04.2015 and numbered 29314, will be complied with regarding transportation, storage and final disposal. Domestic solid wastes originating from the workers at the facility will be collected by the Tavşanlı Municipality on a daily basis.

Hazardous waste is likely to be generated during the assembly and operation phases within the project area. Oil change and maintenance of vehicles and machines to be used within the scope of the project will be done at fuel stations. Contaminated parts, cloths and gloves that may occur

during the construction phase and the hazardous wastes that will arise from the administrative office, maintenance - repair units during the activity in the project area will be collected separately from each other in the Hazardous waste storage area to be built within the project area, not exceeding 180 days, and will be sent to licensed companies.

Noise Pollution

In Article 22 of the Regulation on the Evaluation and Management of Environmental Noise, criteria regarding the level of noise emitted from industrial facilities, workshops, workshops and workplaces have been determined. In this context, clause a of the relevant article states that "The noise level emitted from each industrial facility to the environment cannot exceed the limit values given in Table-4 in Annex-VII of this Regulation". In this table, it is seen that the limit threshold of noise is 70 dB for industries.

The closest settlement to the project Area is Derbent Village and it is 1250 m away from the project site. The noise level of the facility is less than the threshold limits. Therefore, the settlements shall not be affected by noise pollution.

2.4 Public Comments

This Project will be open for public comment on the Verra website after being listed on the pipeline.

2.5 AFOLU-Specific Safeguards

N/A.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Project Activity applies AMS.III.D.- "Methane Recovery in animal manure management systems" (version 21.0).

Applied tools are listed below:

- "Project and leakage emissions from anaerobic digesters" (version 02.0)
- "Project Emissions from Flaring" (version 04.0);
- "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0);
- "Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion" Version 03.0

Moreover, the Project Activity applies AMS.I.D.- “Grid-connected renewable electricity generation” (version 18.0). Sectoral scopes are listed below:

- “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” (version 03.0)
- “Tool to calculate the emission factor for an electricity system” (version 07.0)
- “Tool to determine the remaining lifetime of equipment” (version 01.0)
- “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1

3.2 Applicability of Methodology

Demonstration of applicability conditions for the CDM baseline and monitoring methodology AMS-III.D. (version 21.0), and all above-referred CDM methodological tools are included in the tables below:

Methodology applicability	The proposed project activity
AMS-III.D., "Methane Recovery in animal manure management systems "version 21.0	
This methodology is only applicable under the following conditions: (a) The livestock population in the farm is managed under confined conditions; (b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise “AMS-III.H Methane recovery in wastewater treatment” shall be applied; (c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5 °C; (d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m; (e) No methane recovery and destruction by flaring or	(a) The livestock population of the farms are under confined conditions. Moreover, these conditions are audited by the state since these conditions are mandatory by regulations. (b) Manure or the streams obtained after treatment is not discharged into natural water resources. (c) The annual average temperature of Kütahya, where the project is located, is 21.7 °C. (d) Before the project activity, anaerobic lagoons were used. The depth of the lagoons was more than 1 m. Moreover, the retention time was more than 1 month. (e) In the baseline, anaerobic lagoons were used. Therefore, no flaring and combustion for gainful use take place.

combustion for gainful use takes place in the baseline scenario.	
The project activity shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured; (b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared; (c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when remove	(a) Residual waste from animal manure management system will be used as fertilizer. Since fertilizers are used aerobically, no methane emission will occur. (b) Produced biogas will be burnt in order to generate electricity. In emergency situations, there is a flare to ensure all biogas is burnt. Therefore, the project complies with this condition. (c) The cattle barn and BPP are in same area, and chicken litter is brought from nearby farms. The manure will be taken into digesters just shortly after it is brought. Therefore, storage time will not exceed 45 days.
Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion".	The project does not include methane from landfill, wastewater treatment, or composting of animal manure, or co-digestion of animal manure and other organic matters. Therefore, none of the mentioned methodologies is applicable.

Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	The electricity produced by the project activity will be delivered to the grid.
New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".	The Project Activity does not include any capacity additions
The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	The project is greenfield project, so there is no replaced equipment.
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	Expected emission reduction of the project is 40,257 tCO ₂ e/year which is less than the project. Therefore, the project complies with the condition.

the applicability conditions of the tools referred by the applied methodology AMS-III.D. are explained in the below table.

Methodology applicability	The proposed project activity
AMS-III.D., "Methane Recovery in animal manure management systems "version 21.0	
Project and leakage emissions from anaerobic digesters	The project activity includes the treatment of a biodegradable fraction of animal manure through anaerobic digestion in an anaerobic digester. The project activity meets the applicability condition of the methodological tool.
Project emissions from flaring	There is flaring system in the project to be used in emergency. Therefore, the tool is applicable.
Tool to calculate baseline, project and/or leakage emissions from electricity	The tool is applied for calculating emissions from electricity consumption. For the Project activity energy consumption is achieved from the grid only

consumption	and no captive power plant(s) is/are installed at the site of electricity consumption. Hence, the tool is applicable.
“Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” Version 03.0	This tool is not used because the project emissions results from the auxiliary equipment use will be negligible compared to the project emissions from electricity consumption or flaring.

Demonstration of applicability conditions for the CDM baseline and monitoring methodology AMS-I.D. (version 18.0), and all CDM methodological tools at section 3.1 are included in the tables below:

Methodology applicability	The proposed project activity
AMS-I.D., "Grid-connected renewable electricity generation "version 18.0	
Install a Greenfield plant	The Project is a Greenfield project involving the setting up of a new power plant at a site where there were no power plants in the pre-project scenario. Therefore, the project implies the condition.
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:	Proposed project is a biogas so this condition is not applicable.
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The Project Activity doesn't have any non-renewable component. The Project involves electricity generation by utilizing biogas produced from animal manure. Hence, the condition is not applicable.
Combined heat and power (cogeneration) systems are not eligible under this category.	The project activity is not a co-generation system, i.e.: the simultaneous generation of heat and electrical energy in one process. It consists in a biogas genset unit that produce electricity. The genset heat is recovered in a heat exchanger that transfer the heat to water that is utilized to maintain the required temperature conditions in the mesophilic reactors.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the	The Project Activity does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility

added capacity of the units added by the Project should be lower than 15 MW and should be physically distinct from the existing units.	since there was no renewable energy generation in the project site prior to project activity.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated, or replacement power plant/unit shall not exceed the limit of 15 MW.	The Project Activity is not a retrofit, rehabilitation or replacement project.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration, other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The recovered methane is used for electricity generation for supply to a grid. The baseline for the electricity component is in accordance with the procedure prescribed under this methodology. The recovered methane is not used for heat generation or cogeneration.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The Project Activity uses animal manure as a fuel which does not include any dedicated plantations.

the applicability conditions of the tools referred by the applied methodology AMS-I.D. are explained in the below table.

Methodology applicability	The proposed project activity
AMS-I.D., "Grid-connected renewable electricity generation" version 18.0	
"Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion" Version 03.0	This tool is not used because the project emissions results from the auxiliary equipment use will be negligible compared to the project emissions from electricity consumption or flaring.
"Tool to calculate the emission factor for an electricity system" version 07.0	This tool is not used since the Turkish Ministry of Energy and Natural Resources published the Emission factor applicable for the national grid for wind, solar and other renewable projects.
"Tool to determine the remaining lifetime of equipment" Version 01	Project technical lifetime is determined through " Tool to determine the remaining lifetime of

	equipment” (Version 01). The project option is selected as (a) as below; (a) Use manufacturer’s information on the technical lifetime of equipment and compare to the date of first commissioning; The Project is designed for an operational lifetime of 30 years.
“Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1	This tool is not used due to the fact that the Project Activity is not at the renewal of the crediting period.

3.3 Project Boundary

As noted above, the baseline and monitoring methodologies applied to the proposed project activity are AMS-III.D. “Methane recovery in animal manure management systems”, Version 21.0. and AMS-I.D. “Grid-connected renewable electricity generation” Version 18.0.

In accordance with these Methodologies, the project boundary includes:

- The livestock;
- Animal manure management systems (including centralized manure treatment plant);
- Facilities which recover and flare/combust or use methane;
- The project power plant and all power plants connected physically to the electricity system that the proposed project power plant is connected to.

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted
	CH ₄	Yes	The major source of emissions in the baseline
	N ₂ O	No	Excluded for simplification. This is conservative.

Source	Gas	Included?	Justification/Explanation
Emissions from electricity consumption/generation	CO ₂	Yes	Electricity is consumed from the grid in the baseline scenario
	CH ₄	No	Excluded for simplification. This is conservative.
	N ₂ O	No	Excluded for simplification. This is conservative.
Project	Physical leakage of biogas	CO ₂	Excluded for simplification. This emission source is assumed to be very small.
		CH ₄	The major source of emissions. Leakage from biogas digester has been considered.
		N ₂ O	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from on-site electricity use	CO ₂	These emissions are accounted for since the project activity consumes the electricity from the grid.
		CH ₄	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from the flaring or combustion of gas stream	CO ₂	Flare system will be in use only emergency. If there is any flaring, emission reduction will be excluded during crediting period.
		CH ₄	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from incremental transportation distances	CO ₂	May be an important emission source. Hence, they are accounted
		CH ₄	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from storage of manure	CO ₂	Excluded for simplification. This emission source is assumed to be very small.
		CH ₄	Storage time will be so short. Therefore, This emission source is assumed to be very small.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.

Schematic demonstration of the project boundary also shown in figure below.

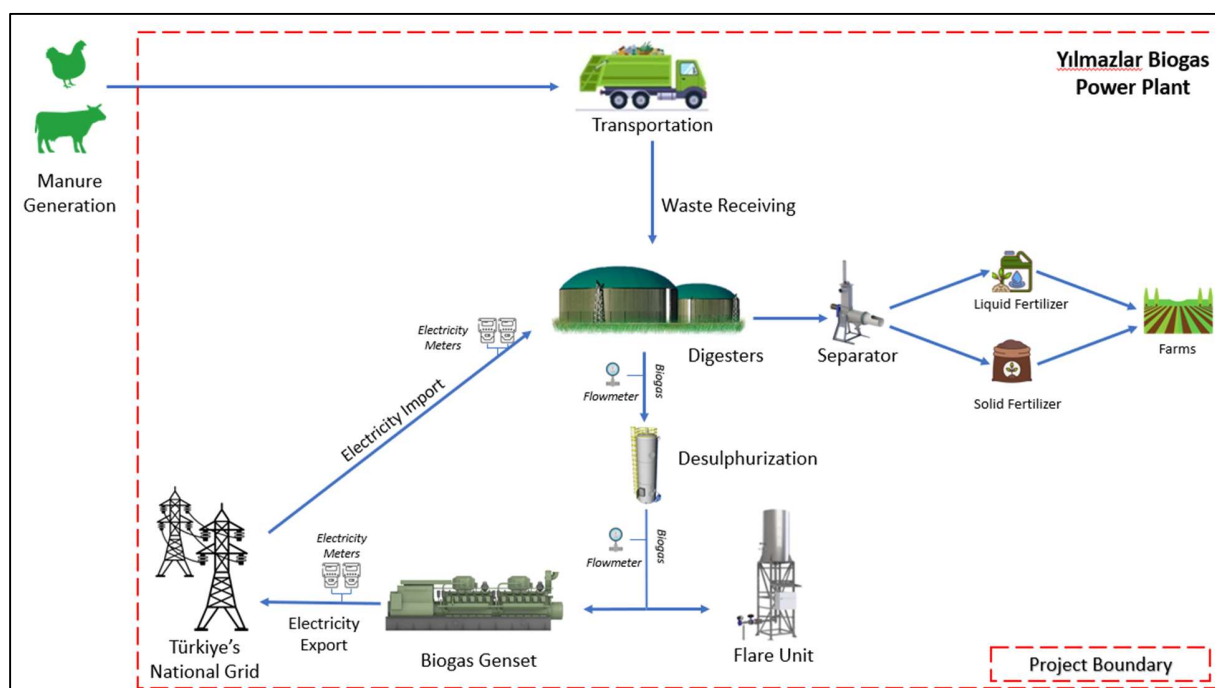


Figure 2 Schematic Demonstration of the Project Boundary

3.4 Baseline Scenario

The proposed project activity has two major GHG emission reduction components. The first one is the reduction of methane emissions resulting from improved animal manure management, whilst the second one is the use of biogas recovered from improved animal manure management for generating power and feeding the electricity generated directly to the national electricity grid network. Assessment of baseline scenario for the both components are provided below:

As per AMS-III.D version 21.0

In line with the applied methodology AMS-III.D. version 21.0, New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the General guidelines for

SSC CDM methodologies version 23.1⁶. The Guidelines describes “procedure to determine baseline scenario for Type II and Type III Greenfield projects” which is a 4-step process to be followed in order to determine whether the requirements are met⁷. Steps to be followed is listed below.

Step 1

Identify the various alternatives available to the project proponent that deliver comparable levels of service, including the proposed project activity or PoA undertaken without being registered as a CDM project activity or PoA.

Step 2

List the alternatives identified in Step 1 that are in compliance with local regulations. If any of the identified baselines is not in compliance with local regulations, then exclude that alternative from further consideration).

Step 3

Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the “Guidelines on the demonstration of additionality of small-scale project activities”.

Step 4

The project activity or PoA is eligible under the methodology if only one alternative remains that:

(a) Is not the proposed project activity or PoA without being undertaken as registered CDM project activity or PoA; and

(b) Corresponds to one of the baseline scenarios provided in the methodology.

If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline

Step 1 – Identify the various alternatives to the project proponent that deliver comparable levels of service, including the proposed project activity undertaken without being registered as a CDM project activity.

The project activity is a greenfield project. According to the applied methodology for greenfield projects, the baseline scenario is selected from the complete set of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Volume 4, Chapter 10, Table 10.14)⁸.

⁶ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210211212225226/MethSSC_Guid25ver23.1.pdf

⁷ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210211212225226/MethSSC_Guid25ver23.1.pdf

⁸ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_10_Ch10_Livestock.pdf

Manure management systems are listed below:

- Pasture, Range and Paddock
- Solid Storage
- Liquid/Slurry Storage
- Anaerobic Lagoons
- Pit Storage
- Dry Lot
- Daily Spread
- Anaerobic Digestion – Biogas
- Burned for Fuel

Step 2 - List the alternatives identified in Step 1 that are in compliance with local regulations. If any of the identified baselines is not in compliance with local regulations, then exclude that alternative from further consideration).

Information for the alternatives are given in the table below.

IPCC Alternatives	Justification
Pasture, Range and Paddock	In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Solid Storage	Solid storage system is not in line with current “Regulation on Livestock Farms” entered into force in 2006.
Liquid/Slurry Storage	This alternative does not comply with the local regulations since “Air Quality Regulation of Turkey” states that liquid wastes shall be stored in impervious storage areas and transferred with closed containers.
Anaerobic Lagoons	In accordance with “Regulation on Livestock Farms” entered into force in 2006, animal manure shall be kept in lagoons made in concrete so that it shall be leakage-proof. Therefore, the alternative complies with the local regulations.

Pit Storage	Pit storage system was compliant with old regulations, but it is not allowed anymore. Animal farms with pit storage that was implemented before the new regulation, are asked to renew their storage system.
Dry Lot	In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Daily Spread	In accordance with “Application Communique on Animal By-Products Regulation Not Used for Human Consumption”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Anaerobic Digestion - Biogas	Anaerobic digestion with biogas would associate with high investment costs and skilled manpower. There is also no legal requirement for aerobic treatment.
Burned for Fuel	Burning animal manure is not allowed in accordance with local regulations.

Step 3: Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the “Guidelines on the demonstration of additionality of small-scale project activities”.

Alternative Scenario	Justification
The manure is disposed in an uncovered anaerobic lagoon;	This alternative is considered a realistic alternative since there is no regulation that restricts the scenario and applicable. The manure is disposed in an uncovered anaerobic lagoon. This is also the real baseline scenario.
Anaerobic Digestion - Biogas	This alternative is proposed project activity. Although there are no regulatory restrictions to implement this scenario, the alternative is

	eliminated due to technological and investment barriers.
--	--

Step 4: If only one alternative remains that is not the proposed project activity undertaken without being registered as a CDM project activity and also corresponds to one of the baseline scenarios provided in the methodology, then the project activity is eligible under the methodology. If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline.

As a result of steps above, only one alternative scenario maintains all the requirements as baseline scenario. That is why, the proposed project activity shall be eligible under AMS-III.D. methodology.

As per AMS-I.D version 18.0

In accordance with paragraph 19 of AMS-I.D. v.18, *“The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.”*

Since the project will generate renewable energy by utilization of biogas and supply it to national grid which is mainly driven by fossil fuel fired power plants, it will reduce CO₂ emissions that would have been generated without the project activity.

3.5 Additionality

Section 3.13.1. of VCS Standard, v4.1, clearly mandates that *“Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project”*. Also, same section of paragraph 2 of the VCS Standard V4.1 states that *“the project proponent shall demonstrate to the validation/verification body that the simplified procedure is appropriate to apply to the project considering the project characteristics, including the context in which the project activity takes place.”*

The paragraph 15 of the approved methodology of AMS-III.D., version 21.0, applied to the proposed project activity, states that *“Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the ‘Guidelines on the demonstration of additionality of small-scale project activities’”*.

As noted in the section 3.4, there is no legal regulation in Turkey that requires the collection and destruction of methane from livestock manure. Thus, it is not required to apply the *“Guidelines on the demonstration of additionality of small-scale project activities”*. Accordingly, Type III component of the proposed project activity, as methane recovery from animal manure, shall be considered as deemed additional.

Moreover, the paragraph 16 of AMS-III.D., version 21.0. clarifies that *“This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this*

methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component”.

Within the boundaries of the Type III component of the proposed project activity, the project captures and utilizes biogas from animal manure in anaerobic digesters to generate electricity with biogas engines/units with the total installed capacity that is lower than 5 MW. Hence, as per the paragraph 16 of AMS-III.D., version 21.0, it is not required to apply the ‘*Guidelines on the demonstration of additionality of small-scale project activities*’ to the Type I component of the proposed project activity which is grid-connected renewable electricity generation. Accordingly, Type I component of the proposed project activity, as grid-connected renewable electricity generation, shall also be considered as deemed additional.

Therefore, the proposed project activity, as per the approved methodology of AMS-III.D., version 21.0, is deemed to be additional without further documents of barriers as per “*Guidelines on the demonstration of additionality of small- scale project activities*”.

3.6 Methodology Deviations

There are no methodology deviations applied during this monitoring period.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emission calculations according to AMS.III.D (version 21.0):

Before the project activity, the manure would decay anaerobically. Therefore, option (a) presented in AMS-III.D., version 21.0 paragraph 17 is utilised. Since there are both cattle and poultry farms that generates manure, emissions from both farms are considered separately and added into calculation.

In accordance with the option(b) baseline emissions are determined based on directly measured quantity of manure and its specific volatile solids content, as follows:

$$BE_y = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j}$$

Where;

BE_y	=	Baseline emissions in year y (t CO ₂ e)
GWP_{CH_4}	=	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period (t CO ₂ e/t CH ₄)
D_{CH_4}	=	CH ₄ density (0.00067 t/m ³ at room temperature (20 °C) and 1 atm pressure)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
MCF_j	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j
$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m ³ CH ₄ /kg-dm)
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)

$MS\%_{BI,j}$ = Fraction of manure handled in baseline animal manure management system j

UF_b = Model correction factor to account for model uncertainties (0.94)

Yılmazlar biogas power plant receives waste from 12 different farms. List of the farms and information related to farms are given in the table below:

Name of the Farm	Type of animal	Number of Animals	Average Weight of Animal (kg)	Rate of received manure
Kayı Farm	Dairy Cattle	2,500	600	100%
As Tarım	Chicken Layer	50,000	1.1	50%
Baş Tarım	Chicken Layer	220,000	1.1	50%
Özkantar Gıda	Chicken Layer	95,000	1.1	50%
Halil Yeşil Yumurta Üretim	Chicken Layer	12,600	1.1	50%
Yoncalı Gıda	Chicken Layer	350,000	1.1	50%
Yumurtalar Tarım	Chicken Layer	28,000	1.1	50%
Emre Kayner	Chicken Layer	18,500	1.1	50%
Halil Küçük	Chicken Layer	15,000	1.1	50%
Saydam Tarım	Chicken Layer	22,000	1.1	50%
Bayraktar	Chicken Layer	330,000	1.1	50%
Ali Aydın	Chicken Layer	70,000	1.1	100%

$$BE_{Kayı\ Farm,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.24 \times 2500 \times 2499.52 \times 1)=19,306\ tCO_2e/y$$

$$BE_{As\ Tarım,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 50000 \times 7.7745 \times 0.5)=976\ tCO_2e/y$$

$$BE_{Baş\ Tarım,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 220000 \times 7.7745 \times 0.5)=4,294\ tCO_2e/y$$

$$BE_{Özkantar\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 95000 \times 7.7745 \times 0.5)=1,854\ tCO_2e/y$$

$$BE_{Halil\ Yeşil\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 12600 \times 7.7745 \times 0.5)=246\ tCO_2e/y$$

$$BE_{Yoncalı\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 350000 \times 7.7745 \times 0.5)=6,831\ tCO_2e/y$$

$$BE_{Yumurtaları\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 28000 \times 7.7745 \times 0.5)=546\ tCO_2e/y$$

$$BE_{Emre\ Kayneri\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 18500 \times 7.7745 \times 0.5)=361\ tCO_2e/y$$

$$BE_{Halil\ Küçük\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 15000 \times 7.7745 \times 0.5)=293\ tCO_2e/y$$

$$BE_{Saydam\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 22000 \times 7.7745 \times 0.5)=429\ tCO_2e/y$$

$$BE_{Bayraktar\ ,y}=28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 330000 \times 7.7745 \times 0.5)=6,440\ tCO_2e/y$$

$$BE_{Ali\ Aydin,y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 70000 \times 7.7745 \times 1) = 2,732 \text{ tCO}_2\text{e/y}$$

As a result, **BE_y = 44,307 tCO₂e/y**

Baseline emission calculations according to AMS.I.D (version 18.0):

In accordance with this methodology, baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where;

BE_y	=	<i>Baseline emissions in year y (t CO₂)</i>
EG_{PJ,y}	=	<i>Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)</i>
EF_{grid,y}	=	<i>Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO₂/MWh)</i>

The project is greenfield project so, in accordance with the paragraph 26 of AMS-I.D., the equation below shall be used for calculation of EG_{PJ,y}:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

EG_{PJ,facility,y}	=	<i>Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)</i>
-----------------------------------	---	--

Turkey's National Electric Grid Emission Factor for the year of 2020 was used. Publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors, for the relevant year with usage of the IPCC's Clean Development Methodology Tool 07-V07.0. For this calculation, information regarding used data set is given below in detail;

- TEİAŞ Turkey's electricity generation-consumption and loss statistics,

- Common prepared report under Turkey's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS) and Verified Carbon Standard (VCS) for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V3.0, Power plant efficiency figures are used

According to this publication Building Margin and Operating Margins below⁹;

- Operating Margin-OM; 0,7424 tCO₂/MWh
- Build Margin-BM; 0,3680 tCO₂/MWh

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where:

$EF_{grid,BM,y}$	=	<i>Build margin CO₂ emission factor in year y (t CO₂/MWh)</i>
$EF_{grid,OM,y}$		<i>Operating margin CO₂ emission factor in year y (t CO₂/MWh)</i>
w_{OM}		<i>Weighting of operating margin emissions factor (per cent)</i>
w_{BM}		<i>Weighting of build margin emissions factor (per cent)</i>

According to CDM Tool 07 “Tool to calculate emission factor for an electricity system” v07.0¹⁰, paragraph 86(b) w_{OM} and w_{BM} for the first crediting period are given below:

⁹

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%99Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/Bform2020.pdf>

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

$$W_{OM} = 0.5$$

$$W_{BM} = 0.5$$

As a result:

$$EF_{grid,CM,y} = 0.7424 \times 0.5 + 0.3680 \times 0.5 = 0.5552 \text{ tCO}_2/\text{MWh}$$

So annual estimated emission reduction,

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y} = 11,700 \times 0.5552 = 6,495 \text{ tCO}_2/\text{year}$$

4.2 Project Emissions

Project Emissions as per AMS-III.D.

The project emissions in year y are calculated for each alternative waste treatment process implemented in the project activity as follows:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where:

PE_y	=	Project emissions in year y (t CO ₂ e)
$PE_{PL,y}$	=	Emissions due to physical leakage of biogas in year y (t CO ₂ e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the biogas stream in the year y (t CO ₂ e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO ₂ e)
$PE_{transport,y}$	=	Emissions from incremental transportation in the year y (t CO ₂ e)
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO ₂ e)

Project emissions from anaerobic digestion ($PE_{PL,y}$)

$PE_{PL,y}$ is calculated according to “TOOL14: Project and leakage emissions from anaerobic digesters” as per AMS-III.D paragraph 21(b) stated.

As per paragraph 23 of the “TOOL14: Project and leakage emissions from anaerobic digesters”, The project emission associated with the anaerobic digester ($PE_{PL,y}$) equals to ($PE_{CH_4,y}$) and is determined as follows:

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Where:

$PE_{CH_4,y}$	=	Project emissions associated with the anaerobic digester in year y (t CO ₂ e)
$Q_{CH_4,y}$	=	Quantity of methane produced in the anaerobic digester in year y (t CH ₄)
$EF_{CH_4,default}$	=	Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO ₂ e)
GWP_{CH_4}	=	Project emissions of methane from the anaerobic digester in year y (t CO ₂ e)

And $Q_{CH_4,y}$ is calculated as follows:

$$Q_{CH_4,y} = Q_{biogas,y} \times f_{CH_4,default} \times \rho_{CH_4}$$

$Q_{CH_4,y}$	=	Quantity of methane produced in the digester in year y (t CH ₄)
$Q_{biogas,y}$	=	Amount of biogas collected at the digester outlet in year y (Nm ³ biogas)
$f_{CH_4,default}$	=	Default value for the fraction of methane in the biogas (m ³ CH ₄ / m ³ biogas)
ρ_{CH_4}	=	Density of methane at normal conditions (t CH ₄ / Nm ³ CH ₄)

Parameter	Value
$EF_{CH_4,default}$	0.028
GWP_{CH_4}	28
$f_{CH_4,default}$	0.6
ρ_{CH_4}	0.00067 t/m ³

Determination of project emissions from flaring of biogas ($PE_{flare,y}$)

In accordance with paragraph 22 of AMS-III.D, “In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool “Project emissions from flaring”. If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.”

Since the methane is used for energy production in biogas engines, destruction efficiency is taken as 100%.

However, there is also flare unit in the plant which has open flare. The flare is installed for the case of emergency and malfunction in the plant. Thus, $PE_{flare,y}$ is considered as zero. Nevertheless, the monitoring plan of the facility shall measure the quantity of captured methane fed to the flare. If such measurement is positive, the emissions from flaring the biogas shall be determined following the procedure described in the methodological tool project emissions from flaring, version 04.0. To be conservative, efficiency of flaring in flare unit is taken as 0%.

Project emissions from flaring are calculated as the sum of emissions from each minute m , based on the methane mass flow in the residual gas ($F_{CH4,RG,m}$) and the flare efficiency

($\eta_{flare,m}$), as follows:

$$PE_{flare,y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

Where:

$PE_{flare,y}$ Project emissions from flaring of the residual gas in year y (tCO₂e)

$F_{CH4,RG,m}$ Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ Flare efficiency in minute m

Parameter	Value
$\eta_{flare,m}$, biogas engine	100%
$\eta_{flare,m}$, flare unit	0%

Determination of project emissions from electricity consumption ($PE_{EC,y}$)

In accordance with paragraph 23 of AMS-III.D, “Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool “Project and leakage emissions from anaerobic digesters”.

Option 1 is chosen to determine project emissions from electricity consumption. As per Option 1 of Tool 14, $PE_{EC,y}$ shall be calculated using the “Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” version 03.0, where the project emission source j referred to in the tool is the total electricity consumption associated with the anaerobic digestion facility.

According to Paragraph 16 of Tool 05, project emissions from the consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

Where:

- $PE_{EC,y}$ = Project emissions from electricity consumption in year y (t CO₂ / yr)
- $EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
- $EF_{EF,j,y}$ = Emission factor for electricity generation for source j in year y (t CO₂/MWh)
- $TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y
- j = Sources of electricity consumption in the project

However, since the project activity involves electricity generation, it is expected that the plant will generate more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation will be claimed under AMS-I.D. v18.0 and energy consumption will be subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} will be considered only if there is energy consumption without energy generation in monthly basis.

Determination of the emission factor for electricity generation $EF_{EF,j,y}$

According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), for the project in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the

latest approved version of the “Tool to calculate the emission factor for an electricity system” (version 07.0)

For the emission factor, the Turkish Ministry of Energy and Natural Resources publication, which indicates Turkey's National Electric Grid Emission Factor for the year 2020, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors for the relevant year with the usage of the IPCC's Clean Development Methodology Tool 07-v07.0. For this calculation, information regarding the data set is given below in detail;

- TEİAŞ Turkey's electricity generation-consumption and loss statistics,
- Common prepared report under Turkey's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS), Verified Carbon Standard (VCS) and Global Carbon Council for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V03.0, Power plant efficiency figures are used

According to this publication;

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh
- Combined Margin-CM (other renewables than solar and wind); 0.5552 tCO₂/MWh¹¹

Determination of project emissions from incremental transportation in the year (PE_{transport})

In accordance with AMS-III.D., version 21.0., PE_{transp.y} is calculated as per the paragraph 14 of AMS-III.AO, version 01.0., which states project emissions due to incremental transport distances (PE_{transp.y}) are calculated based on the incremental distances between:

$$PE_{transp,y} = (Q_y / CT_y) * DAF_w * EF_{CO2,transport} + (Q_{res-waste,y} / CT_{res-waste,y}) * DAF_{res-waste} * EF_{CO2,transport}$$

Where:

Q_y	=	Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)
CT_y	=	Average truck capacity for transportation (tonnes/truck)
DAF_w	=	Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)
$EF_{CO_2,transport}$	=	CO_2 emission factor from fuel use due to transportation (kg CO_2 /km, IPCC default values or local values may be used)
$Q_{res,waste,y}$	=	Quantity of residual waste produced in year y (tonnes)
$CT_{res-waste,y}$	=	Average truck capacity for residual waste transportation (tonnes/truck)
$DAF_{res-waste}$	=	Average distance for residual waste transportation (km/truck)

The distances between farms and Yılmazlar BPP is given in the table below.

Name of the Farm	Distance to the plant (km)
Kayı Farm	0
As Tarım	18
Baş Tarım	25
Özkantar Gıda	25
Halil Yeşil Yumurta Üretim	25
Yoncalı Gıda	23
Yumurtalar Tarım	22
Emre Kayner	24
Halil Küçük	24
Saydam Tarım	24
Bayraktar	24
Ali Aydın	100

In accordance with studies poultry can defecate 3-4% of its weight in a day¹². So, for the calculation 4% has taken to be conservative and Q_y has found accordingly.

Project determine emissions from the storage of manure

Since the storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester in the proposed project activity, $PE_{storage,y}$ equals to zero.

¹² <https://dergipark.org.tr/tr/pub/gbad/issue/29696/319526>

Project Emissions as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO_2 emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the “Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion”. Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed project activity, PE_y as per AMS-I.D., version 18.0., shall be accounted as 0.

4.3 Leakage

In accordance with AMS-III.D. the leakage emission of the project is calculated following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0 and formula in the tool is given below:

$$LE_{AD,y} = LE_{storage,y} + LE_{comp,y}$$

Where:

- $LE_{AD,y}$ = Project emissions from electricity consumption in year y (t CO_2 / yr)
- $LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO_2e)
- $LE_{comp,y}$ = Leakage emissions associated with composting digestate in year y (t CO_2e)

Since there is no storage and no composting in the facility $LE_{AD,y}$ equals to zero.

As per AMS-I.D. version 18.0, “General guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues.”. Since this is not applicable for proposed project, leakage emission resulted from AMS-I.D. v.18 equals to zero.

As a result of evaluations above, leakage emission of proposed project activity is 0.

4.4 Estimated Net GHG Emission Reductions and Removals

Net GHG Emission Reduction and Removals as per AMS-III.D

To calculate the emission reductions as per AMS-III.D, following equation shall be applied as per paragraph 27 of the methodology:

$$ER_{y,ex\ post} = \min[(BE_{y,ex\ post} - PE_{y,ex\ post}), (MD_y - PE_{power,y,ex\ post})]$$

Where:

$ER_{y,ex\ post}$	=	Emission reductions achieved by the project activity based on monitored values for year y (t CO ₂ e)
$BE_{y,ex\ post}$	=	Baseline emissions calculated using equation 1 (for projects using option in paragraph 17(a)) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used
$PE_{y,ex\ post}$	=	Project emissions calculated using equation 6 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_{I, All}$, $Q_{res\ waste,y}$ and if applicable $VS_{LT,y}$
MD_y	=	Methane captured and destroyed or used gainfully by the project activity in year y (t CO ₂ e)
$PE_{power,y,ex\ post}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO ₂ e)

Calculation of methane flared or combusted (MD_y) can only be calculated for verification and shall be calculated as follows:

$$MD_y = BG_{burnt,y} \times w_{CH_4,y} \times D_{CH_4} \times FE \times GWP_{CH_4}$$

Where:

$BG_{burnt,y}$	=	Biogas flared or combusted in year y (m ³)
$w_{CH_4,y}$	=	Methane content in biogas in the year y (volume fraction)
FE	=	Flare efficiency in the year y (fraction)

If flared or combusted biogas data is not available, alternatively, biogas flared or combusted (MD_y) also can be calculated as follows:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \times GWP_{CH_4}$$

Where:

- EG_y = Total electricity generated from the recovered biogas in year y (MWh)
- 3600 = Conversion factor (1 MWh = 3600 MJ)
- NCV_{CH_4} = NCV of methane (MJ/Nm³) (use default value: 35.9 MJ/Nm³)
- EE_y = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:
- Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.
 - Default efficiency of 40 %.

Summary of emission reduction as per AMS-III.D is given in the table below:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	22,700	5,402	0	17,298
2022	44,307	10,545	0	33,762
2023	44,307	10,545	0	33,762
2024	44,307	10,545	0	33,762
2025	44,307	10,545	0	33,762
2026	44,307	10,545	0	33,762
2027	44,307	10,545	0	33,762
2028	21,607	5,142	0	16,465

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Total	310,149	73,814	0	236,335
Average over the Crediting Period	44,307	10,545	0	33,762

Net GHG Emission Reduction and Removals as per AMS-I.D

In accordance with AMS-I.D., net emission reduction and removals has calculated by following formula:

$$ER_y = BE_y - PE_y - LE_y$$

ER_y = Emission reductions in year y (tCO₂)

BE_y = Baseline Emissions in year y (tCO₂)

PE_y = Project emissions in year y (tCO₂)

LE_y = Leakage emissions in year y (tCO₂)

As calculated in section 4.1 baseline emissions BE found as follows:

$$BE_y = EG_{pj,y} * EF_{grid,CM,y} = 11,700 \times 0.5552 = 6,495 \text{ tCO}_2/\text{year}$$

In accordance with paragraph 39 of AMS-I.D, project emission (PE_y) equals to zero and also leakage emission (LE_y) was assumed as 0.

As a result

$$ER_y = BE_y = 6,495 \text{ tCO}_2/\text{year}$$

Summary of emission reduction as per AMS-I.D is given in the table below:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	3,328	0	0	3,328

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2022	6,495	0	0	6,495
2023	6,495	0	0	6,495
2024	6,495	0	0	6,495
2025	6,495	0	0	6,495
2026	6,495	0	0	6,495
2027	6,495	0	0	6,495
2028	3,167	0	0	3,167
Total	45,465	0	0	45,465
Average over the Crediting Period	6,495	0	0	6,495

Summary of net emission reductions as per AMS-III.D and AMS-I.D are given in the table below.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	26,028	5,402	0	20,626
2022	50,802	10,545	0	40,257
2023	50,802	10,545	0	40,257
2024	50,802	10,545	0	40,257
2025	50,802	10,545	0	40,257

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2026	50,802	10,545	0	40,257
2027	50,802	10,545	0	40,257
2028	24,774	5,142	0	19,632
Total	355,614	73,814	0	281,800
Average over the Crediting Period	50,802	10,545	0	40,257

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP _{CH₄}
Data unit	t CO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC Fifth Assessment Report: Climate Change 2013 ⁸⁶
Value applied:	28
Justification of choice of data or description of measurement methods and procedures applied	Default value IPCC Fifth Assessment Report (AR5)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	ρ _{CH₄}
Data unit	t CH ₄ / m ³ CH ₄
Description	Density of methane at normal conditions
Source of data	‘Thermophysical properties of fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane’ by B.A. Younglove, J.F. Ely Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Value applied:	0.00067
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0

Purpose of Data	Calculation of baseline/project emissions
Comments	corrected to reference conditions

Data / Parameter	UF _b
Data unit	Dimensionless
Description	Model correction factor to account for model uncertainties (
Source of data	CDM Methodology AMS-III.D -Methane recovery in animal manure management systems
Value applied:	0.94
Justification of choice of data or description of measurement methods and procedures applied	Default values given in AMS-III.D.
Purpose of Data	Calculation of baseline/project emissions
Comments	-

Data / Parameter	MCF _j
Data unit	-
Description	Methane conversion factor for the baseline AWMS _j
Source of data	Obtained from 2019 IPCC Refinement, Table 10.17, p. 67 (uncovered anaerobic lagoon for 17.9°C, warm temperate moist)
Value applied:	0.73
Justification of choice of data or description of measurement methods and procedures applied	Default value. Uncovered anaerobic lagoon, Value for 17.9°C.

Purpose of Data	Used in project emission/baseline calculations
Comments	A conservativeness factor should be applied by multiplying MCF values (estimated as per the above bullet) with a value of 0.94 to account for the 20 per cent uncertainty in the MCF values as reported by IPCC

Data / Parameter	B _{0,LT}
Data unit	m ³ CH ₄ /kg-dm
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	Obtained from 2019 IPCC Refinement, Table 10.16,
Value applied:	0.24 for dairy cattle 0.39 for chicken layer
Justification of choice of data or description of measurement methods and procedures applied	Default values for maximum methane producing capacity
Purpose of Data	Used in project emission/baseline calculations
Comments	-

Data / Parameter	VS _{LT,y}
Data unit	kg -dm/animal/year
Description	The default value for the volatile solid excretion per day on a dry-matter basis for a defined livestock population
Source of data	Elaborated from 2019 IPCC Refinement, Table 10.13A p. 10.55 Middle East
Value applied:	2,499.52 dairy cattle

	7.77 for chicken layer
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in project emission/baseline calculations
Comments	-

Data / Parameter	$EF_{CH_4, default}$
Data unit	t CH ₄ leaked / t CH ₄ produced
Description	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester
Source of data	IPCC (2006), Flesch et al. (2011) and Kurup (2003)
Value applied:	0.028
Justification of choice of data or description of measurement methods and procedures applied	As per Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0 Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction shall use the default of 0.028
Purpose of Data	Calculation of project emissions
Comments	Applicable to the step "Determination of project emissions of methane from the anaerobic digester"

Data / Parameter	$f_{CH_4, default}$
Data unit	m ³ CH ₄ / m ³ biogas corrected to reference conditions ²
Description	Default value for the fraction of methane in the biogas

Source of data	The default value was derived based on reported values from registered projects and research papers (Davidsson, 2007)
Value applied:	0.6
Justification of choice of data or description of measurement methods and procedures applied	As per Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0, 0.6 is the default value.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Republic of Turkey Ministry of Energy in Emission Factor 2020
Value applied:	0.5552
Justification of choice of data or description of measurement methods and procedures applied	Calculated ex ante and fixed for the 7-year renewable crediting period.
Purpose of Data	Determination of baseline emissions
Comments	-

Data / Parameter	EF _{CO₂,transport}
Data unit	kgCO ₂ /km
Description	CO ₂ emission factor from fuel use due to transportation

Source of data	Project and leakage emissions from road transportation of freight v.1.0						
Description of measurement methods and procedures applied	Distance between manure producer and the plant via Map applications						
Frequency of monitoring/recording	Annually						
Value applied:	<table border="1"> <thead> <tr> <th>Vehicle Class</th><th>Emission factor</th></tr> </thead> <tbody> <tr> <td>Light Vehicle</td><td>0.245</td></tr> <tr> <td>Heavy Vehicle</td><td>0.129</td></tr> </tbody> </table>	Vehicle Class	Emission factor	Light Vehicle	0.245	Heavy Vehicle	0.129
Vehicle Class	Emission factor						
Light Vehicle	0.245						
Heavy Vehicle	0.129						
Monitoring equipment	-						
QA/QC procedures applied	Default value						
Purpose of data	Project emission calculations						
Calculation method	-						
Comments	-						

Data / Parameter	$TDL_{j,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j or k in year y
Source of data	Methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0)
Value applied:	3%
Justification of choice of data or description of measurement methods and procedures applied	According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), the project belongs to the case of Scenario A, use as default values of 3% for project emission

	since electricity consumption by all project electricity consumption sources is smaller than the electricity consumption of all baseline electricity consumption sources, and 3% for baseline emissions.
Purpose of Data	Calculation of project emissions from electricity consumption
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meters / EPIAS records
Description of measurement methods and procedures applied	This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid. In case it is calculated then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid
Frequency of monitoring/recording	Monthly
Value applied:	Electricity generation estimated as 11,700 MWh ¹³ (<i>ex-ante value, will be monitored ex post</i>)
Monitoring equipment	Electricity meters

¹³ Generation License

QA/QC procedures applied	Electricity meters are subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing to ensure accuracy.
Purpose of data	This parameter is required for calculating baseline emissions from electricity generation due to waste treatment under the project activity.
Calculation method	No calculation is needed measurement values are used.
Comments	-

Data / Parameter	N _{da,y}
Data unit	Number
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponents and farms
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Monthly
Value applied:	365
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	Number of animal for type LT are kept same on average in accordance with facility's capacity and plan. In other words, mortality and growth rate is kept same. Therefore, no growth or deduction is expected in normal circumstances. Average number of animals are provided by statement of each facility.

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Project proponents
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Monthly
Value applied:	2500 diary cattle, 1,211,100 chicken layers <i>(as per agreements signed with farmers)</i>
Monitoring equipment	-
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	Number of animal for type LT are kept same on average in accordance with facility's capacity and plan. In other words, mortality and growth rate is kept same. Therefore, no growth or deduction is expected in normal circumstances. Average number of animals are provided by statement of each facility.

Data / Parameter	$MS\%_{BI,j}$
Data unit	%
Description	Fraction of manure handled in system j in the project activity
Source of data	Project proponent

Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Annually
Value applied:	100% for Kayı Farm and Ali Aydın farm; for the rest, 50% of the produced manure treated by the proposed project activity. (as per agreements signed with farmers)
Monitoring equipment	-
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	W_{site}
Data unit	kg
Description	Average animal weight of a defined livestock population at the project site
Source of data	Plant Logs
Description of measurement methods and procedures applied	Average weight for each type of animal.
Frequency of monitoring/recording	-
Value applied:	640 kg for dairy cattle 1.5 kg for chicken layer
Monitoring equipment	-

QA/QC procedures applied	
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y
Source of data	Flow meters and SCADA system of the plant
Description of measurement methods and procedures applied	The amount of biogas recovered and fueled, flared or used gainfully shall be monitored ex post, using flow meters.
Frequency of monitoring/recording	Annually, based on continuous flow measurement with accumulated volume recording (e.g. hourly/daily accumulated reading).
Value applied:	-
Monitoring equipment	Flow meters
QA/QC procedures applied	Calibration of the flow meters as per manufacturer specifications and regulations
Purpose of data	Emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	Q_{biogas}
------------------	--------------

Data unit	Nm ³ biogas
Description	Amount of biogas collected at the digester outlet in year y
Source of data	Flow meter
Description of measurement methods and procedures applied	The volumetric flow is measured by flow meters and data is shown in SCADA and recorded by SCADA operator daily.
Frequency of monitoring/recording	Continuously measurement by the flow meter. Data to be aggregated monthly and yearly
Value applied:	3,704,429 Nm ³ (ex-ante value, will be monitored ex-post)
Monitoring equipment	Flow meter
QA/QC procedures applied	-
Purpose of data	Project emissions calculations
Calculation method	-
Comments	For the validation, the used formula is given below: $Q_{biogas} = \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}) / f_{CH4,default}$ Actual value will be monitored ex post.

Data / Parameter	W _{CH4}
Data unit	%
Description	Methane content in biogas in the year y
Source of data	Analyser
Description of measurement methods and procedures applied	The fraction of methane in the biogas should be measured with a continuous analyser

Frequency of monitoring/recording	Same frequency as the flow or, with periodical measurements at a 90/10 confidence/precision level by following the “Standard for sampling and surveys for CDM project activities and Programme of Activities”,
Value applied:	60% (<i>ex-ante value, will be monitored ex-post</i>)
Monitoring equipment	Analyser
QA/QC procedures applied	-
Purpose of data	Emission reductions and project emissions calculations
Calculation method	-
Comments	-

Data / Parameter	FE
Data unit	%
Description	Flare efficiency
Source of data	-
Description of measurement methods and procedures applied	As per para.22 of the applied methodology, if the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. Flare unit will be used only emergency situations. Efficiency of flare unit is assumed to be 0% to be conservative. In other words, methane reduction from flare unit will not be claimed.
Frequency of monitoring/recording	-
Value applied:	0% for flare unit

	100% for biogas engine
Monitoring equipment	-
QA/QC procedures applied	Regular maintenance will be carried out according with manufacturer recommendation to ensure optimal flare operation. It will be ensured that the temperature of the flare ($T_{EG,m}$) and the flow rate of the residual gas to the flare ($F_{RG,m}$) is within the manufacturer`s specification for the flare ($SPEC_{flare}$).
Purpose of data	Project emissions calculations
Calculation method	-
Comments	-

Data / Parameter	Q_y and $Q_{res\ waste,y}$
Data unit	Tons
Description	Quantity manure brought to the plant
Source of data	On-site records
Description of measurement methods and procedures applied	On-site data sheets recorded monthly using weigh bridge.
Frequency of monitoring/recording	Monthly
Value applied:	-
Monitoring equipment	Weight bridge
QA/QC procedures applied	Weighbridge will be subject to periodic calibration (in accordance with stipulation of the weighbridge supplier)
Purpose of data	Project emission calculations
Calculation method	-

Comments	For the validation, amount of waste per farm was calculated based on animal number and potential excretion rate by animal weight. In accordance with a study performed by the Ministry of Agriculture and Forestry ¹⁴ daily excretion rate is 4% of animal weight.
----------	---

Data / Parameter	CT _y and CT _{res waste,y}
Data unit	Tons/truck
Description	Average truck capacity for transportation
Source of data	On-site measurement
Description of measurement methods and procedures applied	On-site data sheets recorded monthly using weigh bridge.
Frequency of monitoring/recording	Monthly
Value applied:	24 tons/truck in average
Monitoring equipment	Weight bridge
QA/QC procedures applied	Weighbridge will be subject to periodic calibration (in accordance with stipulation of the weighbridge supplier)
Purpose of data	Project emission calculations
Calculation method	-
Comments	-

Data / Parameter	DAF _w and DAF _{res waste}
------------------	---

14

[https://arastirma.tarimorman.gov.tr/tavukculuk/Belgeler/web%20English%20Doc/journal%20\(Dergimiz\)/Dergimiz%20Cilt%2011%20Sayi%201/Cilt%2011%20Sayi%201%20Makale%204%20Yeni%20Bir%20Teknoloji%20ile%20Kurutulan%20Tavuk%20D%20C4%B1%C5%9Fk%C4%B1s%C4%B1n%C4%B1n.pdf](https://arastirma.tarimorman.gov.tr/tavukculuk/Belgeler/web%20English%20Doc/journal%20(Dergimiz)/Dergimiz%20Cilt%2011%20Sayi%201/Cilt%2011%20Sayi%201%20Makale%204%20Yeni%20Bir%20Teknoloji%20ile%20Kurutulan%20Tavuk%20D%20C4%B1%C5%9Fk%C4%B1s%C4%B1n%C4%B1n.pdf)

Data unit	km/truck
Description	Average incremental distance for raw solid or product transportation
Source of data	Map applicaitons
Description of measurement methods and procedures applied	Distance between manure producer and the plant via Map applications
Frequency of monitoring/recording	Annually
Value applied:	Please see section 4.2
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Project emission calculations
Calculation method	-
Comments	-

5.3 Monitoring Plan

The monitoring methodology is based on directly measuring the biogas produced in the digesters, the biogas sent to the flare unit, and the electricity generated in the engines, as shown in the table below. The monitoring plan continuously measures the quantity and quality of biogas generated and flared. The main variables that need to be determined are the quantity of biogas burnt or gainfully used on a dry basis in year y ($BG_{burnt,y}$) and the quantity of methane used to generate electricity. The methodology also measures the energy generated using biogas ($EG_{PJ,y}$).

From the monitoring methodology, it could be seen that there are the following main variables to be measured:

Table 2: Summary of Monitoring Plan

Parameter	Description
$BG_{burnt,y}$	Burnt biogas volume in year y (m^3)

Q_{biogas}	Amount of biogas collected at the digester outlet in year y (Nm ³ biogas)
$F_{\text{CH}_4, \text{RG}, y}$	Amount of methane which is sent to the flare unit in year y
$FE_{\text{flare}, m, \text{biogas engine}}$	Flare efficiency of biogas engine (%)
W_{CH_4}	Methane fraction in the biogas (%)
$EG_{\text{PJ}, y}$	Net electricity delivered to the grid (MWh)

Responsibilities for data processing and management lie with Yilmazlar. Therefore, they established a VER team. This team is responsible for monitoring all data required to estimate emission reductions. Life Enerji also assists VER Team concerning the monitoring aspects of the Project. The plant manager has the main responsibility for collecting and archiving the data. According to the monitoring plan, qualified technicians monitor and record the data. All the technicians receive proper training to ensure that they understand their specific tasks and equipment handling. The records are double-checked by the General Manager of the Proposed Project, who is responsible for the accuracy and frequency of the measurements.

EPIAS stores the power generation data. The website of EPIAS is accessible by the Project owner with their unique user ID and password. Once accessed, the Project owners can retrieve electricity generation and consumption data. The Project owner uses the same data for invoicing purposes. The electricity generation data are reported on a monthly basis.

Data Processing and collection

The monitoring equipment and other plant equipment are connected to a Programmable Logic Controller (PLC), which collects all the monitoring data and sends them to the SCADA system so that the operator can read the main parameters in real-time. A SCADA operator continuously read the data and record the read parameters. The main source for biogas data is the Project activity's PLC-SCADA system. The system automatically reads measurement devices and the data manually recorded. This data is used to calculate the total emission reductions.

1) Flow measurements

Gas flow is measured by flow meters, including the flow into the flare, generators and the total flow. These parameters are required to be normalised at 0°C and 1 atm, using actual gas temperature and pressure data to be measured with temperature and pressure transmitters. These in-built transmitters are integrated into each flow meter and equipped with a flow computer so that the values sent to the SCADA system have already been normalised (converted to normal cubic meters). Therefore, temperature and pressure data at each flowmeter do not need to be separately monitored. Methane content in the gas is measured by a gas analyser connected to the PLC-SCADA system.

Calibration: The flow sensors are calibrated according to the gas's specified temperature, pressure, and composition as per the manufacturer's recommendation. The equipment selected will allow dynamic compensation for these parameters, normalised to standard temperature, pressure, and gas composition. There will be a periodic verification according to the requirement of equipment specifications.

2) Gas Quality and efficiency of the flare

An enclosed flare unit determines methane destruction with an inbuilt flame detection unit. The project emissions from flaring and emission reductions from the destruction of biogas have been monitored at this point. The readings have been taken and recorded internally by compatible and proper measurement devices with high data quality. The PLC system automatically converts the measured parameters to dry- basis and normal conditions. The instruments monitor parameters continuously. The emissions are calculated using the methodological tool "Project emissions from flaring" (v04.0).

The concentration of methane and oxygen in the gaseous stream and the exhaust gas of the flare are the parameters that are essential for the calculation of emission reductions and the safe and efficient operation of the system.

The concentration of methane and oxygen in the biogas stream is controlled by a common sample line installed in the main collection system piping and measured continuously by two separate analysers to measure methane, oxygen and carbon dioxide each. The sensors are designed to operate within specified temperature and pressure conditions.

The concentration of methane and oxygen in the exhaust gas stream is monitored by a common sample line installed in the upper section of the flare.

Calibration: Analysers are periodically calibrated according to the manufacturers and regulations on "Metering and Testing of Metering Systems"¹⁵ of the Ministry of Industry and Technology recommendation. Calibration equipment will provide +/- 1% accuracy by volume.

3) Electricity Generation:

The electricity generation data is stored by EPIAS, the financial settlement centre of TEIAS (the national grid operator). The website of EPIAS is accessible by the Project owner with their ID used for control and checks and invoicing purposes. The electricity generation data is reported on a monthly basis. The procedure involves the following tasks:

- Accessing the website of EPIAS
- Obtaining the electricity generation and consumption reports
- Store the data in electronic version on the computer of Yilmazlar on the plant
- Issue invoices to related distribution company based on the data published in EPIAS.

¹⁵ See, http://www.sanayi.gov.tr/download/osgm/olcu_aletleri_muayene_yonetmelik.zip (page 2)

A Backup Unit for Systematic Storage

i. Electricity Generation:

There are two measurement devices; a primary and a secondary (i.e. backup or check-meter) measurement device for quality assurance, to be used if the primary meter fails. Both the main and backup measurement devices measure and store the aggregate total electricity import and export for the whole Project in real-time. The electricity meters are under the responsibility of distribution company, and they are located in the sub-station, which also belongs to the distribution company.

The invoices are kept by the Project owner as hard copies. Furthermore, the EPIAS system stores the reports electronically, which is accessible to the Project owner whenever necessary. After the monthly reports are obtained from EPIAS, data are aggregated using a separate spreadsheet. The monthly generation and consumption data are entered into the spreadsheet, which calculates the emission reductions during the monitoring period.

ii. Methane Destruction:

Data Storage: The electronic monitoring system (PLC) will periodically read the data continuously and recorded manually by an operator. Written documents (e.g. equipment replacement protocols, accident logs, maintenance records, back-ups etc.) will be kept safely. Electronically backup of the data will be conducted on a daily basis. A hard copy backup of all relevant data will be printed out monthly. Calibration records for all instrumentation will be constantly collected and archived. All data and records required for verification are kept for two years after the project crediting period ends or the last issuance of VERs, whichever is later.

Calibration Procedures:

Maintenance and calibration of measurement devices: According to manufacturer specifications, all measurement devices will be purchased and maintained as specified in this report. All measurement devices that are used in monitoring will be subject to a quality control procedure that will include regular maintenance and calibration in agreement with legal and/or manufacturer requirements. According to the relevant regulation, "The Turkish Electricity Market Regulation Agency (EPDK) sets rules on the accuracy of electricity meters that are used by power plants feeding into the grid. The rules are part of the EPDK regulation 25056 from 22/03/2003¹⁶. The table in Article 11 of the regulation specifies electricity meters of the accuracy class 0.5 for power plants between less than 10 MW for and refers to compliance with the International Electrotechnical Commission's norm EN 60687. TEIAS, whose employees visit the plant for the meter readings monthly, is in charge of ensuring adherence to these rules. Calibration and maintenance procedures follow the requirements."

If any equipment malfunction or breakdown occurs, corrective actions are carried out in a timely manner to minimise the risk of emissions that are not intended. To ensure data quality,

¹⁶ <https://www.epdk.gov.tr/Detay/Icerik/3-0-77-3/tebligler>

operational staff will be trained appropriately to take action effectively in such cases. The plant operator periodically inspects the plant to check visually if there are any apparent problems. In case of any findings, these are documented.

In case of an organisational change within the Project Plant, a qualified person will be assigned to carry out and manage the monitoring and verification procedures. In case of fire, earthquake or another similar emergency situation, the Project owner's data as backup stores will be used to maintain conservativeness with specific procedures. For extraordinary events where such backup data are unavailable, the amount of biogas consumed will be calculated using the power generation figures and engines' full-load electrical efficiencies by using conservative values and other assumptions to maintain conservativeness. Regarding the project emissions from flaring, data recording and aggregation and procedures for extraordinary events are the same as described above (i.e. the methane destruction component).

Monitoring will be carried out following the procedures set by the applicable methodology and all applicable tools of the above-explained parameters. The monitoring consists of:

Responsibilities for monitoring

The project owner is responsible for the installed equipment operation and maintenance. The project owner is also responsible for the administration of the data, setting up a VER team who will be responsible for monitoring all data required to estimate emission reductions.

Registration of the monitored data

For Yilmazlar BPP Project, VER-Team will be responsible for monitoring equipment for quality assurance. The data measured by the monitoring equipment will be stored and will be processed into a monitoring report, which the project owner will submit. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team will regularly check the monitoring system for errors. In the case of errors, corrective actions will be undertaken by the VER-Team, or if required, by the monitoring equipment supplier.

QA/QC procedure

Strong quality assurance and quality control procedures will be taken to monitor the equipment and data collection. Equipment and facilities will be subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or correct data reading, the data will be cross-checked with other parameters and data leading to the most conservative emission reductions will be considered for calculating the emission reductions.

Measuring of exported and imported electricity

The main meter for monitoring the electricity import and export is at the grid connection. Monthly protocols based on the readings confirmed by TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed project activity. The electricity meters are bi-directional and will measure both the imported and exported electricity from and to the grid.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Value applied:	Will be provided after listing
Comments	-

Data / Parameter	$N_{da,y}$
Data unit	days
Description	Number of days animal is alive in the farm in the year y
Value applied:	365
Comments	-

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y

Value applied:	2500 diary cattle, 1,211,100 chicken layers
Comments	-

Data / Parameter	W_{site}
Data unit	kg
Description	Average animal weight of a defined livestock population at the project site
Value applied:	640 for diary cattle, 1.5 for chicken layer
Comments	-

Data / Parameter	$MS\%_{BI,j}$
Data unit	%
Description	Fraction of manure handled in system j in the project activity
Value applied:	100% for Kayı Farm and Ali Aydın farm, For the rest, 50% of the produced manure treated by the proposed project activity.
Monitoring equipment	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y

Value applied:	Will be provided after listing
Comments	-

Data / Parameter	Q_{biogas}
Data unit	Nm ³ biogas
Description	Amount of biogas collected at the digester outlet in year y
Value applied:	Will be provided after listing
Comments	

Data / Parameter	W_{CH_4}
Data unit	%
Description	Methane content in biogas in the year y
Value applied:	Will be provided after listing
Monitoring equipment	Analyser
Comments	-

Data / Parameter	FE
Data unit	%
Description	Flare efficiency
Value applied:	0% for flare unit 100% for biogas engine
Comments	-

Data / Parameter	Q_y and $Q_{res\ waste,y}$
Data unit	Tons
Description	Quantity manure brought to the plant
Value applied:	Will be provided after listing
Comments	-

6.2 Baseline Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.3 Project Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.4 Leakage

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.5 Net GHG Emission Reductions and Removals

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.